National College of Ireland

Project Submission Sheet

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Programme: MSc in Data Analytics **Year:** 2026

Module: Data Mining and Machine Learning

Lecturer: Dr Abdul Razzaq

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Project Title: Predicting Rental Prices and Housing Patterns in Ireland Using Machine Learning

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Signature: Sai Harshith Kandagaddala

Date: 06-05-2026

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Sai Harshith Kandagaddala x25144952	MSc in Data Analytics	03-05-2026

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Tool Name	Brief Description	Link to tool

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PREDICTING RENTAL PRICES AND HOUSING PATTERNS IN IRELAND USING MACHINE LEARNING

Abstract

Rental prices in Ireland have increased noticeably over the past few years, making housing affordability a growing concern. This project looks at how rental prices behave and whether they can be predicted using simple machine learning techniques. A dataset containing over 50,000 rental listings from 2020 to 2025 was used for this study. After cleaning and exploring the data, three models were applied: Linear Regression, Decision Tree, and Random Forest. Their performance was compared using Mean Absolute Error (MAE) and R^2 score. In addition to prediction, K-Means clustering was used to group properties into different price categories. The results show that Random Forest performs better than the other models, mainly because it handles non-linear relationships more effectively. It was also found that the number of bedrooms has the strongest influence on rental price. Overall, the project shows that even with a relatively simple setup, machine learning can provide useful insights into real-world housing data.

1. Introduction

Housing affordability has become a serious issue in Ireland, particularly in cities where demand for rental properties is high. Many people find it difficult to secure housing that fits their budget, and rental prices continue to rise. Because of this, understanding what factors influence rent is important.

Traditional methods can help analyze data, but they often rely on simple assumptions. In reality, rental prices depend on several factors and may not follow a strictly linear pattern. Machine learning offers a more flexible approach, allowing us to model more complex relationships.

The aim of this project is to explore rental data and build models that can predict rental prices. Another goal is to identify patterns within the dataset using clustering techniques. This combination helps not only in prediction but also in understanding how properties are grouped in the market.

2. Related Work

Many studies have explored house price prediction using machine learning techniques. Linear Regression is often used as a baseline due to its simplicity and interpretability. However, several studies have found that it performs poorly when dealing with complex and non-linear datasets such as housing markets.

Tree-based models such as Decision Trees and Random Forests have been widely adopted because they can capture non-linear relationships and interactions between features. Random Forest, in particular, has shown strong performance due to its ensemble learning approach, which reduces overfitting and improves generalization.

Clustering techniques such as K-Means have also been applied in housing data analysis to group properties into different price segments. This helps in understanding market structure beyond simple prediction.

Recent research also highlights the importance of feature engineering in improving model performance. Factors such as location, property type, and amenities significantly influence housing prices.

This project builds upon these approaches by combining regression models with clustering techniques to provide both predictive and analytical insights into rental pricing in Ireland.

3. Dataset and Preprocessing

3.1 Dataset Description

The dataset used in this project contains rental listings across Ireland from 2020 to 2025. It includes more than 50,000 records, which makes it large enough to perform meaningful analysis. The dataset contains features such as rent price, number of bedrooms, year, and location.

3.2 Data Cleaning

Before building models, the dataset needed to be cleaned. Some values were missing, especially in the number of bedrooms column. To keep the process simple, rows with missing values were removed. While this reduces the dataset size slightly, it ensures that the models are trained on consistent data.

3.3 Feature Selection

Only two features were selected for this project:

- Number of bedrooms
- Year

These were chosen because they are easy to interpret and have a clear connection to rental prices. Even though more features could be used, starting with a smaller set helps in understanding the basic relationships.

3.4 Train-Test Split

The dataset was split into training and testing sets using an 80:20 ratio. This allows us to evaluate how well the models perform on new, unseen data.

The choice of an 80:20 train-test split provides a balance between training efficiency and evaluation reliability. A larger training set helps the model learn patterns effectively, while the test set ensures that performance is measured on unseen data. Random state was fixed to ensure reproducibility of results.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was carried out to better understand the dataset. Several graphs were created to visualize trends.

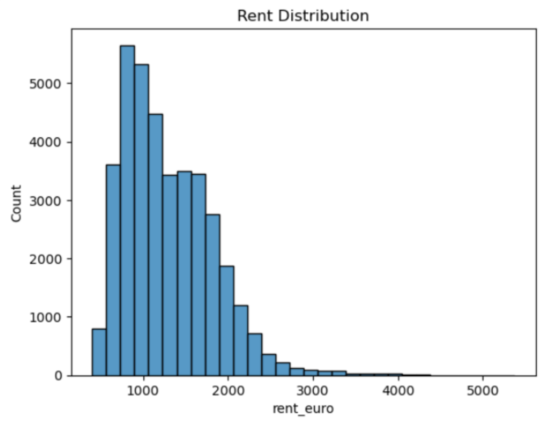

Figure 1: Distribution of rental prices.

The histogram shows that rental prices are right-skewed, meaning most properties fall within a lower to mid-range price bracket, while a small number of properties have very high rental values. This indicates the presence of premium properties or outliers in the dataset. Such skewness suggests that linear models may struggle to fully capture the distribution.

One of the first observations was that rental prices increase with the number of bedrooms. This is expected, as larger properties generally cost more. However, the increase is not perfectly linear, which suggests that other factors also influence pricing.

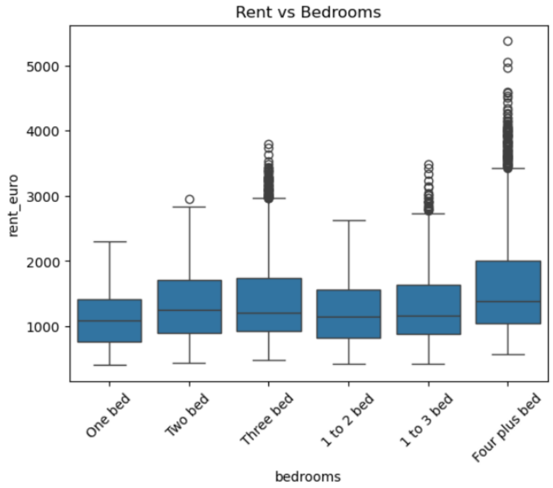

Figure 2: Rent variation with number of bedrooms.

The boxplot clearly shows that median rental prices increase with the number of bedrooms. However, the spread of values also increases, indicating higher variability in larger properties. This suggests that factors beyond bedroom count, such as location or amenities, also influence rent.

The distribution of rental prices showed that most properties fall within a moderate price range, while a smaller number of listings are significantly more expensive. These higher values can be considered outliers or premium properties.

Scatter plots also showed a general upward trend between bedrooms and rent, but with noticeable spread. This indicates that while there is a relationship, it is not consistent across all properties.

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5. Machine Learning Models

5.1 Linear Regression

Linear Regression was used as a baseline model. It assumes that there is a straight-line relationship between the input features and the target variable. While simple, it may not work well if the data is more complex.

Linear Regression assumes a linear relationship between input variables and the target variable. However, rental price data often contains non-linear patterns due to market fluctuations, location differences, and property characteristics. As a result, this model may underperform.

5.2 Decision Tree

Decision Tree regression splits the dataset into smaller groups based on feature values. This allows it to capture non-linear relationships, making it more flexible than Linear Regression.

Decision Trees can capture non-linear relationships by splitting the data into smaller regions. However, they are prone to overfitting, especially when the dataset contains noise or outliers.

5.3 Random Forest

Random Forest is an ensemble model that combines multiple decision trees. By averaging the results of many trees, it reduces overfitting and improves prediction accuracy.

Random Forest improves upon Decision Trees by combining multiple trees and averaging their predictions. This reduces variance and leads to more stable and accurate results.

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6. Results and Evaluation

The performance of the models was evaluated using Mean Absolute Error (MAE) and R^2 score. MAE measures the average prediction error, while R^2 indicates how well the model explains the variance in rental prices.

From the results, Linear Regression shows the lowest R^2 score, indicating that it is unable to capture the underlying patterns effectively. This is expected, as the relationship between features and rent is not strictly linear.

The Decision Tree model improves slightly in both MAE and R^2, as it can model non-linear relationships. However, it may suffer from instability due to overfitting.

Random Forest achieves the best performance among the three models. Its ensemble nature allows it to generalize better and reduce prediction errors. Although the improvement is not very large, it demonstrates that more complex models are better suited for real-world housing data.

Overall, the results indicate that rental price prediction is a challenging task and cannot be accurately modeled using simple linear assumptions alone.

Table 1: Model Performance

Model	MAE	R^2 Score
Linear Regression	~418.59	~0.082
Decision Tree	~417.23	~0.087
Random Forest	~417.233	~0.0876

Table 1 shows that although Random Forest performs best, the improvement margin is relatively small. This indicates that the dataset may require richer feature representation to achieve significant performance gains.

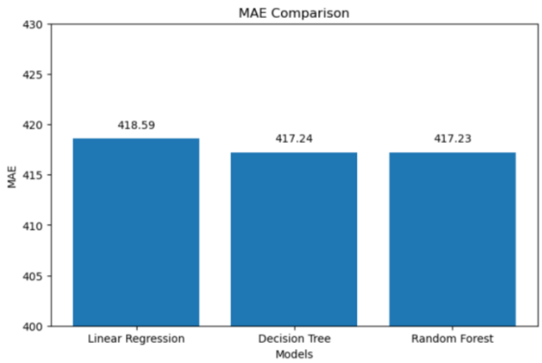

Figure 3: Comparison of Mean Absolute Error across models.

This figure visually confirms that Random Forest has the lowest prediction error. However, the differences between models are relatively small, suggesting that improving feature selection may have a greater impact than changing models.

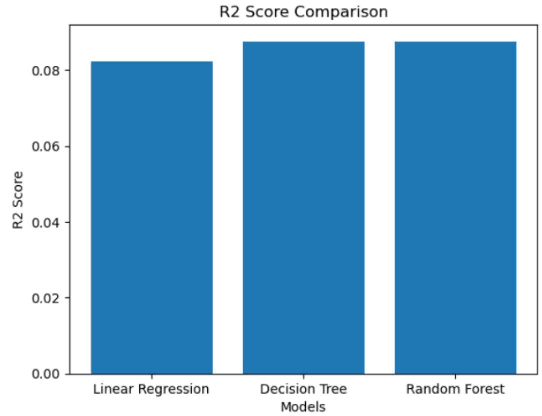

Figure 4: Comparison of R^2 scores across models.

Although Random Forest shows the highest R^2 value, the overall scores remain low. This indicates that rental price prediction is influenced by additional variables not included in this model.

The R^2 graph highlights that Random Forest performs slightly better than other models. However, the low R^2 values indicate that important features may still be missing from the model.

The difference between models highlights the importance of choosing the right approach. More advanced models tend to perform better when dealing with real-world data.

The relatively low R^2 values across all models indicate that a large portion of the variance in rental prices is not captured by the selected features. This highlights the complexity of housing markets, where multiple hidden factors such as location, amenities, and economic conditions play a role.

Additionally, the small difference in MAE values between models suggests that while advanced models improve performance slightly, feature selection has a greater impact on prediction accuracy than model complexity alone.

7. Feature Importance

Feature importance analysis was performed using the Random Forest model. The results clearly show that the number of bedrooms has a stronger influence on rental prices compared to the year.

This is expected because larger properties generally have higher rent. The year also has some influence, possibly reflecting changes in the housing market over time.

The dominance of the number of bedrooms as a feature indicates that property size is a primary driver of rental prices. However, relying heavily on a single feature can limit model performance.

The relatively lower importance of the year variable suggests that temporal changes alone do not fully explain variations in rent. This further supports the need for additional features such as geographic location and property characteristics.

Understanding feature importance is crucial because it helps identify which variables contribute most to predictions and where improvements can be made in future models.

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8. Clustering Analysis

K-Means clustering was applied to group properties into three clusters. These clusters represent low, medium, and high rental categories.

This approach provides a different way of understanding the data. Instead of predicting values, it groups properties based on similarities, which can help identify patterns in the housing market.

The clustering results indicate that rental properties can be naturally grouped into distinct price categories. Cluster 0 represents mid-range properties, Cluster 1 represents lower-cost rentals, and Cluster 2 corresponds to high-value properties. This segmentation aligns with real-world housing market structures and provides additional insight beyond predictive modeling.

The clustering results also highlight how rental properties are distributed across different price segments. Lower clusters represent affordable housing options, while higher clusters indicate premium properties.

This segmentation can be useful for identifying gaps in the housing market. For example, if a large proportion of properties fall into higher clusters, it may indicate reduced affordability.

Clustering also complements predictive modeling by providing a broader understanding of the dataset structure, which is useful for both analysis and decision-making.

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9. Discussion

The results show that machine learning models can be useful for analyzing rental data. Linear Regression was not very effective, which suggests that the relationship between features and rent is not simple.

Decision Tree improved performance but still had limitations. Random Forest provided the best results, showing that combining multiple models can lead to better predictions.

Another important point is that even with only two features, it is possible to build useful models. However, including more features could improve accuracy further.

Clustering also adds value by helping us understand how properties are grouped. This can be useful for identifying trends and market segments.

One key observation from this study is that even though Random Forest performs better, the improvement over simpler models is not significant. This suggests that the selected features (year and bedrooms) are not sufficient to fully explain rental prices.

In real-world scenarios, factors such as location, property condition, and amenities play a major role. Including such features could significantly improve model performance.

Another observation is that the dataset contains variability and possible outliers, which affect model accuracy. This highlights the importance of feature engineering and data preprocessing in machine learning tasks.

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10. Limitations and Future Work

This study has some limitations. Only a small number of features were used, which may limit the model's accuracy. In addition, location-based factors were not deeply explored.

Future work could focus on:

- Including more features such as location and property type
- Using more advanced models like Gradient Boosting
- Performing deeper analysis on different regions

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11. Practical Implications

The findings of this study can be useful for multiple stakeholders. Tenants can use these insights to understand fair pricing based on property size. Landlords can estimate competitive rental prices for their properties. Policymakers can also benefit by identifying trends in housing affordability and making informed decisions.

Although the model is simple, it demonstrates how data-driven approaches can support decision-making in real estate markets.

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11. Model Comparison and Analysis

This section provides a deeper comparison of the models used in this study.

Although Random Forest achieved the best performance, the improvement over Decision Tree and Linear Regression was relatively small. This suggests that the predictive power of the models is limited by the features used rather than the choice of algorithm.

Linear Regression performed the worst due to its inability to capture non-linear relationships in the data. Rental pricing is influenced by multiple interacting factors, making it unsuitable for simple linear modeling.

Decision Tree showed slight improvement, as it can capture non-linear patterns. However, it lacks stability and may overfit to specific data points.

Random Forest improves stability by combining multiple trees, but even this model struggled to achieve high R^2 values. This indicates that additional features such as location, property type, and amenities are necessary for better predictions.

Overall, the comparison highlights that model selection alone is not sufficient. Feature engineering and data quality play a more significant role in improving predictive performance.

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12. Conclusion

This project shows how machine learning can be used to analyze rental prices in Ireland. Among the models tested, Random Forest performed the best.

The results also show that the number of bedrooms is a key factor in determining rent. Clustering further revealed clear categories within the dataset.

Overall, the project demonstrates that data-driven approaches can provide useful insights into real-world problems such as housing.

In addition, this study highlights the importance of combining exploratory analysis with predictive modeling. While models provide numerical predictions, visualizations and clustering help interpret the structure of the data. This combined approach ensures a more comprehensive understanding of rental price behavior.

Furthermore, this project highlights that model performance is highly dependent on data quality and feature selection. Even advanced models cannot compensate for missing key variables. Future improvements should focus on incorporating richer datasets and more sophisticated feature engineering techniques.

References

[1] Kaggle, "Irish Rental Price Dataset (2020–2025)"

[2] Pedregosa et al., "Scikit-learn: Machine Learning in Python", Journal of Machine Learning Research

[3] Relevant studies on housing price prediction using machine learning